B base V $\longrightarrow$ dim V = n

B' base W $\longrightarrow$ dim W = m.

$$\begin{tikzcd} \mathbb{R}^n \arrow[r, bend left, "(\ )_B^{-1}"] \arrow[rrr, bend right] & V \arrow[r, bend left, "f"] & W \arrow[r, bend left, "(\ )_{B'}"] & \mathbb{R}^m \\ (\bar{v})_B \arrow[r, bend left] \arrow[rrr, bend right] & \bar{v} \arrow[r, bend left] & f(\bar{v}) \arrow[r, bend left] & (f(\bar{v}))_{B'} \end{tikzcd}$$

$M_{BB'}f = (f(\bar{v}))_{B'}$

$f : \mathbb{R}^2 \longrightarrow \mathbb{R}^3 \ / \ f(x,y) = (x+y, 2x-y, x)$

$B = \{(1\,1)\ (-1\,2)\}$ E = base canónica.

$B' = \{(0\,1\,0)\ (0\,0\,2)\ (1\,0\,-1)\}$. E'

Hallar: $M_{BB'}f$, $M_{BE}f$, $M_{EE'}f$

$M_{BB'}f = (f(v_1))_{B'}\ (f(v_2))_{B'}$

$$M_{BB'}f = \left[f\begin{pmatrix}1\\1\end{pmatrix}_{B'}\ f\begin{pmatrix}-1\\2\end{pmatrix}_{B'} \right] = \begin{bmatrix} 1 & -4 \\ 3/2 & 0 \\ 2 & 1 \end{bmatrix}$$

$f(1\,1) = (2,1,1) \longrightarrow (2\,1\,1) = \alpha(0\,1\,0) + \beta(0\,0\,2) + \gamma(1\,0\,-1)$.

$= (\gamma;\ \alpha;\ 2\beta - \gamma)$

$$\begin{array}{ccc|c} \alpha & \beta & \gamma & \\ 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 1 \\ 0 & 2 & -1 & 1 \end{array}$$

$\alpha = 1$

$\beta = 3/2$

$\gamma = 2$.

$f(-1\,2) = (1\,-4\,-1) \longrightarrow (1\,-4\,-1) = \alpha(0\,1\,0) + \beta(0\,0\,2) + \gamma(1\,0\,-1)$

$$\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & -4 \\ 0 & 2 & -1 & -1 \end{array}$$

$\alpha = -4$

$\beta = 0$

$\gamma = 1$

$$M_{BE} f = \left[f\begin{pmatrix}1\\1\end{pmatrix}_E \quad f\begin{pmatrix}-1\\2\end{pmatrix}_E \right] = \begin{bmatrix} 2 & 1 \\ 1 & -3 \\ 1 & -1 \end{bmatrix}$$

$f(11) = (2\,1\,1)$ $\quad f(-12) = (1\,-4\,-1)$

$$M_{EE'} f = \left[f\begin{pmatrix}1\\0\end{pmatrix}_{E'} \quad f\begin{pmatrix}0\\1\end{pmatrix}_{E'} \right] = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ 1 & 0 \end{bmatrix}$$

$E = \{(10)\,(01)\}$.

$f(10) = (1\ 2\ 1)$ $\quad f(01) = (1, -1, 0)$

$B = \{v_1\ v_2\ v_3\ v_4\}$ base V

$B' = \{w_1\ w_2\ w_3\}$ base W

$f : V \longrightarrow W \ \Big|\ M_{BB'} f = \begin{pmatrix} 1 & -1 & 0 & 1 \\ 1 & 1 & 1 & -2 \\ 2 & 0 & 1 & -1 \end{pmatrix}$

$M_{BB'} f = (f(v))_{B'}$

a) $f(0)$, $f(v_2)$, $f(v_1 - 2v_2 + v_4)$

$f(0) = 0$.

$(f(v_2))_{B'} = (-1\ 1\ 0) = (-w_1 + w_2)$

$f(v_1 - 2v_2 + v_4) = f(v_1) - 2f(v_2) + f(v_4)$

$(f(v_1 - 2v_2 + v_4))_{B'} = (f(v_1))_{B'} - 2(f(v_2))_{B'} + (f(v_4))_{B'}$

$= w_1 + w_2 + 2w_3 - 2(-w_1 + w_2) + w_1 - 2w_2 + w_3$.

$= w_1 + w_2 + 2w_3 + 2w_1 - 2w_2 + w_1 - 2w_2 + w_3$

$f(v_1 - 2v_2 + v_4) = 4w_1 - 3w_2 + w_3$.

b) Hallar $\operatorname{Nu} f$ $\operatorname{Im} f$ $f^{-1}(w_1 - w_2)$

$(\operatorname{Im} f)_{B'} = \langle (f(v_1))_{B'}, (f(v_2))_{B'}, (f(v_3))_{B'}, (f(v_4))_{B'} \rangle$

$(1\,1\,2)$ $(-1\,1\,0)$ $(0\,1\,1)$ $(1\,-2\,1)$

$\operatorname{Im} f \langle w_1 + w_2 + 2w_3, -w_1 + w_2 \rangle$

$\mathrm{Nu}\, f \quad \bar{v} \mid f(\bar{v}) = 0 \;\rightarrow\; (f(\bar{v}))_{B'} = 0$

$$(M_{BB'} f)\, \bar{v}_B = 0.$$

$$\begin{pmatrix} 1 & -1 & 0 & 1 \\ 1 & 1 & 1 & -2 \\ 2 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{pmatrix} = 0 \Rightarrow \left(\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 0 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 0 & 1 & -1 & 0 \end{array}\right) \rightarrow \left(\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 0 \\ 0 & 2 & 1 & -3 & 0 \\ \cancel{0} & \cancel{-2} & \cancel{1} & \cancel{-3} & \cancel{0} \end{array}\right)$$

$$\lambda_1 - \lambda_2 + \lambda_4 = 0 \rightarrow \lambda_1 = \lambda_2 - \lambda_4.$$

$$2\lambda_2 + \lambda_3 - 3\lambda_4 = 0. \rightarrow \lambda_3 = -2\lambda_2 + 3\lambda_4.$$

$$(\bar{v})_B = (\lambda_2 - \lambda_4,\ \lambda_2,\ -2\lambda_2 + 3\lambda_4,\ \lambda_4)$$

$$= \lambda_2 (1\ 1\ -2\ 0) + \lambda_4 (-1\ 0\ 3\ 1)$$

$$N f = \langle (1\,1\,-2\,0)\ (-1\,0\,3\,1) \rangle = \langle v_1 + v_2 - 2v_3 ;\ -v_1 + 3v_3 + v_4 \rangle$$

$f^{-1}(w_1 - w_2)$

$$v \in f^{-1}(w_1 - w_2) \Rightarrow f(v) = w_1 - w_2$$

$$(f(v))_{B'} = (w_1 - w_2)_{B'} \Leftrightarrow M_{BB'} f \cdot v_B = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & 0 & 1 \\ 1 & 1 & 1 & -2 \\ 2 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \rightsquigarrow (\bar{v})_B = (\lambda_2 - \lambda_4 + 1,\ \lambda_2,\ -2\lambda_2 + 3\lambda_4 - 2,\ \lambda_4)$$

$$= \lambda_2 (1\,1\,-2\,0) + \lambda_4 (-1\,0\,3\,1) + (1\,0\,-2\,0)$$

$$(f^{-1}(w_1 - w_2))_B = \langle (1\,1\,-2\,0\ (-1\,0\,3\,1) \rangle + (1\,0\,-2\,0)$$

$$f^{-1}(w_1 - w_2) = \langle v_1 + v_2 - 2v_3 ;\ -v_1 + 3v_3 + v_4 \rangle + (v_1 - 2v_3)$$

$f : U \longrightarrow V$ $\quad$ B base U $\quad$ $g \circ f : U \longrightarrow W$.

$g : V \longrightarrow W$. $\quad$ B' base V

B'' base W. $\quad$ Dato $M_{BB'} f \quad M_{B'B''} g$.

$$f + g \longrightarrow g(f)$$

$$(\bar{v})_B \longrightarrow (f(\bar{v}))_{B'} \longrightarrow g\left[(f(\bar{v}))\right]_{B''}$$

$$M_{BB''}(g \circ f) = M_{B'B''} g \cdot M_{BB'} f$$

$$(v)_B \xrightarrow{M_{BB'}f} (f(v))_{B'} \qquad (v)_B \xrightarrow{M_{BB'}(id)} (v)_{B'} \longrightarrow C_{BB'}$$

$f: U \to V$ $\qquad$ B, B' base U $\qquad$ Datos.

$g: V \to W$ $\qquad$ $D\ D'$ base V $\qquad$ $M_{BD}f$

$F\ F'$ base W. $\qquad$ $M_{D'F'}g$

Calcula $M_{B'F'}(g \circ f)$ $\qquad$ $v \xrightarrow{Mf} f(v) \xrightarrow{Mg} g(f(v)) = g \circ f$.

$$(\bar{v})_{B'} \xrightarrow{C_{B'B}} (\bar{v})_B \xrightarrow{M_{BD}f} (f(\bar{v}))_D \xrightarrow{C_{DD'}} (f(\bar{v}))_{D'} \xrightarrow{M_{D'F'}g} g(f(\bar{v}))_{F'} \xrightarrow{C_{F'F}} g(f(\bar{v}))_F$$

$$M_{B'F}(g \circ F) = C_{F'F} \cdot M_{D'F'}g \cdot C_{DD'} \cdot M_{BD}f \cdot C_{BB'}.$$

$B = \{(1\ -1)\ (2\ 0)\}$ $\qquad$ $B' = \{(2\ 0)\ (1\ 3)\}$.

$C_{BB'} = \left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix}_{B'} \begin{pmatrix} 2 \\ 0 \end{pmatrix}_{B'} \right\}$

$(1\ -1) = \alpha(2\ 0) + \beta(1\ 3)$

$1 = 2\alpha + \beta. \rightarrow \alpha = 2/3.$

$-1 = 3\beta \rightarrow \beta = -1/3.$

$(2\ 0) = \alpha(2\ 0) + \beta(1\ 3).$

$2 = 2\alpha + \beta. \rightarrow \alpha = 1.$

$0 = 3\beta \rightarrow \beta = 0$

$C_{BB'} = \begin{pmatrix} 2/3 & 1 \\ -1/3 & 0 \end{pmatrix}$

$\downarrow$

$C_{BE} = \cancel{C_{BB'}} \cdot \begin{pmatrix} 1 & 2 \\ -1 & 0 \end{pmatrix}$

$C_{EB} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}_B \begin{pmatrix} 0 \\ 1 \end{pmatrix}_B \right\}$

$M_{EB}F$

$A \in \mathbb{R}^{n \times n}$ $\quad$ $v \in \mathbb{R}^n$ $\qquad$ v es autovector de A si $Av = \underbrace{\lambda}_{\text{autovalor}} v$.

$\det(A - \lambda I) = 0. \longrightarrow \lambda =$ autovalor.

$Av = \lambda v \longrightarrow (A - \lambda I)v = 0.$

$A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$ Hallar autovector y autovalor.

$\det |A - \lambda I| = 0.$ $\begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{vmatrix} 1-\lambda & 1 \\ 4 & 1-\lambda \end{vmatrix} = (1-\lambda)^2 - 4 = 0.$

$\lambda = -1$ o $\lambda = 3.$

$(A - \lambda I)\bar{v} = 0.$

$\lambda = -1 \rightarrow \left(\begin{array}{cc|c} 2 & 1 & 0 \\ 4 & 2 & 0 \end{array}\right. \rightarrow \left(\begin{array}{cc|c} 2 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right. \rightarrow \begin{array}{l} 2x_1 + x_2 = 0 \\ -2x_1 = x_2 \end{array}$ $S_{\lambda_1} = \langle (1-2) \rangle.$

$\lambda = 3 \rightarrow \left(\begin{array}{cc|c} -2 & 1 & 0 \\ 4 & -2 & 0 \end{array}\right.$ $S_{\lambda=3} = \langle (1,2) \rangle.$

$B = \{ (1-2) \ (12) \}.$

$M_B f = (f(v))_B = \lambda (v)_B.$

$\cancel{Mf = \begin{pmatrix} 1 \\ 2 \end{pmatrix}}$ $Mf = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}$

$M_B f = \left[\left(f\begin{pmatrix} 1 \\ -2 \end{pmatrix}\right)_B ; \left(f\begin{pmatrix} 1 \\ 2 \end{pmatrix}\right)_B \right] = \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}.$

$M_B f = \begin{pmatrix} 3 & -1 & 1 \\ 0 & -5 & 0 \\ 0 & 7 & 2 \end{pmatrix}$ $B = \{ v_1 \ v_2 \ v_3 \}$ Hallar B'

$M_{B'} f$ diagonalizable.

$\det (M_B f - \lambda I) = 0. \Rightarrow \begin{vmatrix} 3-\lambda & -1 & 1 \\ 0 & -5-\lambda & 0 \\ 0 & 7 & 2-\lambda \end{vmatrix} = 0 \rightarrow \lambda = 3$

$\lambda = -5$

$\lambda = 2.$

$(M_B f - \lambda I)\bar{v} = 0$

$\lambda = 3$ $\left(\begin{array}{ccc|c} 0 & -1 & 1 & 0 \\ 0 & -8 & 0 & 0 \\ 0 & 7 & -1 & 0 \end{array}\right.$ $\rightsquigarrow S_{\lambda=3}(M_B f) = \langle (100) \rangle = \langle v_1 \rangle.$

$\lambda = -5 \rightsquigarrow S_{\lambda=5}(M_B f) = \langle (-1-44) \rangle = \langle -v_1 - 4v_2 + 4v_3 \rangle$

$\lambda = 2 \rightsquigarrow S_{\lambda=2}(M_B f) = \langle (101) \rangle = \langle v_1 + v_3 \rangle$

$B' = \{ v_1 ; -v_1 - 4v_2 + 4v_3 , v_1 + v_3 \}.$

$M_{B'} f = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

Dato. $A \in \mathbb{R}^{3\times3}$ $\quad \exists\ C \in \mathbb{R}^{3\times3} \ / \ A = C D C^{-1}$

$C \cdot C^{-1} = I$.

$$C = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \longrightarrow \left(\begin{array}{ccc|ccc} a & b & c & 1 & 0 & 0 \\ d & e & f & 0 & 1 & 0 \\ g & h & i & 0 & 0 & 1 \end{array}\right)$$

$$\underset{A}{C^{-1} A C} = \underbrace{C^{-1} C}_{I} D \underbrace{C^{-1} C}_{I}$$

$A = (\ ) \rightarrow M_E f = A$. $\quad \overset{D}{B} \ / \ M_B f = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix}$.

$$A = C D C^{-1}$$
$$\downarrow \qquad \downarrow$$
$$M_{EE} f \qquad M_{BB} f.$$

$$(v)_E \xrightarrow{C_{EB}} (v)_B. \xrightarrow{M_{BB} f} (f(v))_B \xrightarrow{C_{BE}} (f(v))_E.$$

$$\underset{A}{M_{EE} f} = \underset{C}{C_{BE}} \cdot \underset{D}{M_{BB} f} \cdot \underset{C^{-1}}{C_{EB}}.$$

calcular $A^8 = C \ D \ C^{-1}$

$$D^8 \rightarrow \begin{matrix} \lambda_1^8 & & \\ & \lambda_2^8 & \\ & & \lambda_3^8 \end{matrix}.$$

$z \in \mathbb{C}$ $\quad \bar{z} = z + 4i$ y $\arg(\bar{z}) = \frac{\pi}{4} \rightarrow z^4 = ?$

$$\left.\begin{array}{l} z = a + bi \\ \bar{z} = a - bi \end{array}\right\} \quad \bar{z} - z = 4i$$
$$a - bi - a - bi = 4i.$$
$$-2bi = 4i$$
$$b = 2$$

$\cancel{\arg(\bar{z}) = \sqrt{a^2+b^2}}$.

$$\arg(z) = \operatorname{arctg}\left(\frac{y}{x}\right)$$
$$\Downarrow$$
$$\arg(\bar{z}) = \operatorname{arctg}\left(\frac{-b}{a}\right) = \frac{\pi}{4}$$
$$-\frac{2}{a} = \operatorname{tg}\frac{\pi}{4} \Rightarrow a = \frac{-2}{1} = a = -2.$$

$$z = -2 + 2i \Rightarrow z^2 = (-2+2i)(-2+2i)$$
$$z^2 = 4 - 4i - 4i - 4 = -8i$$
$$z^4 = (-8i)(-8i) = -64$$

$$\| z^4 = -64 \|$$

$$S = \left\{ x \in \mathbb{R}^4 \;\middle|\; \begin{array}{l} x_1 - x_2 + 2x_3 = 0 \\ -x_1 + x_3 + x_4 = 0 \end{array} \right\} \qquad T = \langle (1\,1\,-2\,1)\ (2\,0\,1\,3) \rangle$$

S+T.

$x_1 = x_2 - x_3 \cdot 2$

$x_4 = x_1 - x_3 \longrightarrow x_4 = x_2 - 2x_3 - x_3 \Rightarrow x_4 = x_2 - 3x_3$

$$(x_1\ x_2\ x_3\ x_4) = (x_2 - 2x_3,\ x_2,\ x_3,\ x_2 - 3x_3)$$
$$= x_2(1\,1\,0\,1) + (-2, 0, 1, -3)\, x_3 .$$

$S = \langle (1101)(-201-3) \rangle \qquad T \langle (11-21)(2013) \rangle$

$$\begin{array}{cccc} 1 & 1 & 0 & 1 \\ -2 & 0 & 1 & -3 \\ 1 & 1 & -2 & 1 \\ 2 & 0 & 1 & 3 \end{array} \rightarrow \begin{array}{cccc} 1 & 1 & 0 & 1 \\ -2 & 0 & 1 & -3 \\ 0 & 0 & 2 & 0 \\ \cancel{0} & \cancel{0} & \cancel{2} & \cancel{0} \end{array} \rightarrow \begin{array}{cccc} 1 & 1 & 0 & 1 \\ 0 & 2 & 1 & -1 \\ 0 & 0 & 2 & 0. \end{array} \Rightarrow \text{Dim } 3$$

$\{(1101)(-201-3)(11-21)\}$

$\{(1101)(021-1)(11-21)\}$.